

# Subgrid Terms in the compressible Euler Equation

## 1 The Unfiltered Equations

We begin with the Euler Equations in conservative form:

$$\begin{aligned} \partial_t \rho + \nabla \cdot (\rho \mathbf{v}) &= 0 \\ \partial_t (\rho \mathbf{v}) + \nabla \cdot (\rho \mathbf{v} \mathbf{v} + P \mathbf{l}) &= 0 \\ \partial_t \left( \frac{1}{2} \rho \mathbf{v}^2 + \frac{P}{\gamma - 1} \right) + \nabla \cdot \left[ \left( \frac{1}{2} \rho \mathbf{v}^2 + \frac{\gamma P}{\gamma - 1} \right) \mathbf{v} \right] &= -\dot{\mathcal{E}} \end{aligned} \tag{1}$$

where  $\rho$  is the gas mass density,  $\mathbf{v}$  is the fluid velocity,  $P$  is the thermal pressure,  $\mathbf{l}$  is the identity tensor. The total energy density can be written as

$$E = \frac{P}{\gamma - 1} + \frac{1}{2} \rho |\mathbf{v}|^2$$

where  $\gamma = 5/3$ . Using the ideal gas EOS,

$$P = \frac{\rho k_B T}{\mu m_p}$$

where  $T$  is the gas temperature,  $k_B$  is the Boltzmann constant,  $m_p$  is the mass of the proton, and  $\mu = 0.62$  is the mean molecular weight. The radiative cooling per unit volume is given by

$$\dot{\mathcal{E}} = n^2 \Lambda(T), \quad n = \frac{\rho}{\mu m_p}$$

where  $n$  is the number density and  $\Lambda(T)$  is the cooling function.

## 2 The Filtering Approach

Given a filtering kernel  $G_\Delta(\mathbf{x})$ , The filtering operation can be defined as:

$$\bar{f}(\mathbf{x}, t) = \int G_\Delta(\mathbf{x} - \mathbf{x}') f(\mathbf{x}', t) d^3 x', \quad \tilde{f} = \frac{\rho f}{\bar{\rho}}$$

Where  $\bar{f}$  is the density weighted average or the Favre average.

For a top hat kernel, the filtering can be written as

$$\langle f \rangle = \frac{1}{V_{\text{cell}}} \int f d^3x$$

Germano's generalised central moments for a pair of fields  $a$  and  $b$  is defined as

$$\tau(a, b) = \widetilde{ab} - \widetilde{a}\widetilde{b} = \widetilde{\delta a \delta b}$$

where  $\delta a = a - \widetilde{a}$  is the fluctuations about the mean of the field. By construction,  $\widetilde{\delta a} = 0$ .

For three fields  $a, b$ , and  $c$ , we can write

$$\tau(a, b, c) = \widetilde{abc} - \widetilde{a}\tau(b, c) - \widetilde{b}\tau(c, a) - \widetilde{c}\tau(a, b) - \widetilde{a}\widetilde{b}\widetilde{c}$$

### 3 Coarse Graining the Equations

$$\begin{aligned} \frac{\partial}{\partial t} \langle \rho \rangle + \nabla \cdot \langle \rho \mathbf{v} \rangle &= 0 \\ \frac{\partial}{\partial t} \langle \rho \mathbf{v} \rangle + \nabla \cdot [\langle \rho \mathbf{v} \mathbf{v} \rangle + \langle P \rangle \mathbf{I}] &= 0 \\ \frac{\partial}{\partial t} \left\langle \frac{1}{2} \rho \mathbf{v}^2 + \frac{P}{\gamma - 1} \right\rangle + \nabla \cdot \left\langle \left( \frac{1}{2} \rho \mathbf{v}^2 + \frac{\gamma P}{\gamma - 1} \right) \mathbf{v} \right\rangle &= -\langle n^2 \Lambda(T) \rangle \end{aligned} \quad (2)$$

We can expand the term  $\langle \rho \mathbf{v} \mathbf{v} \rangle$  to get

$$\langle \rho \mathbf{v} \mathbf{v} \rangle = \frac{\langle \rho \mathbf{v} \rangle \langle \rho \mathbf{v} \rangle}{\langle \rho \rangle} + \mathbf{R}_{\rho vv} \quad (3)$$

$\mathbf{R}_{\rho vv}$  is a symmetrical tensor that can be read as the residual momentum-flux tensor. We can also write it in terms of Favre averaged velocities,

$$\mathbf{R}_{\rho vv} = \langle \rho \rangle \widetilde{\mathbf{v} \mathbf{v}} - \widetilde{v} \widetilde{v}$$

Using Germano's central moments, we can write the components of the tensor.

$$\mathbf{R}_{ij} = \langle \rho \rangle \tau(v_i, v_j)$$

We can expand the term  $\langle \rho \mathbf{v}^2 / 2 \rangle$  as

$$\left\langle \frac{\rho \mathbf{v}^2}{2} \right\rangle = \text{Tr} \left\langle \frac{\rho \mathbf{v} \mathbf{v}}{2} \right\rangle$$

We can decompose this with the relation in Equation 3.

$$\left\langle \frac{\rho \mathbf{v}^2}{2} \right\rangle = \frac{1}{2} \text{Tr} \left\langle \frac{\langle \rho \mathbf{v} \rangle \langle \rho \mathbf{v} \rangle}{\langle \rho \rangle} \right\rangle + \frac{1}{2} \text{Tr} (\mathbf{R}_{\rho vv})$$

We can use this decomposition to find pressure from the total energy.

$$\frac{\langle P \rangle}{\gamma - 1} = \langle E \rangle - \left[ \frac{1}{2} \text{Tr} \left\langle \frac{\langle \rho \mathbf{v} \rangle \langle \rho \mathbf{v} \rangle}{\langle \rho \rangle} \right\rangle + \frac{1}{2} \text{Tr} (R_{\rho vv}) \right] \quad (4)$$

### 💡 The Subgrid Momentum Flux is Invariant to Galilean Transform

**Claim:** Under a Galilean boost  $\mathbf{v} \rightarrow \mathbf{v}' = \mathbf{v} + \mathbf{V}$  (constant  $\mathbf{V}$ , with  $\mathbf{x}' = \mathbf{x} + \mathbf{V}t$ ), the subgrid terms  $R_{\rho vv}$ , is unchanged.

**Proof:** Under the boost, the scalars  $\rho$ ,  $P$ ,  $T$  are unchanged, so  $\langle \rho \rangle' = \langle \rho \rangle$  and  $\langle P \rangle' = \langle P \rangle$ . The momentum transforms as

$$\langle \rho \mathbf{v} \rangle' = \langle \rho (\mathbf{v} + \mathbf{V}) \rangle = \langle \rho \mathbf{v} \rangle + \mathbf{V} \langle \rho \rangle.$$

Define the resolved velocity  $\tilde{\mathbf{v}} \equiv \langle \rho \mathbf{v} \rangle / \langle \rho \rangle$ , so that  $\tilde{\mathbf{v}}' = \tilde{\mathbf{v}} + \mathbf{V}$ . From the definition,

$$R_{\rho vv} = \langle \rho \mathbf{v} \mathbf{v} \rangle - \langle \rho \rangle \tilde{\mathbf{v}} \tilde{\mathbf{v}}.$$

Boosting the first term:

$$\langle \rho \mathbf{v}' \mathbf{v}' \rangle = \langle \rho \mathbf{v} \mathbf{v} \rangle + \mathbf{V} \langle \rho \mathbf{v} \rangle + \langle \rho \mathbf{v} \rangle \mathbf{V} + \langle \rho \rangle \mathbf{V} \mathbf{V}.$$

Boosting the second term:

$$\langle \rho \rangle \tilde{\mathbf{v}}' \tilde{\mathbf{v}}' = \langle \rho \rangle \tilde{\mathbf{v}} \tilde{\mathbf{v}} + \mathbf{V} \langle \rho \mathbf{v} \rangle + \langle \rho \mathbf{v} \rangle \mathbf{V} + \langle \rho \rangle \mathbf{V} \mathbf{V}.$$

The cross terms and the  $\mathbf{V} \mathbf{V}$  term are identical and cancel in the difference, so

$$R'_{\rho vv} = R_{\rho vv}.$$

We can also define a similar Energy Flux term  $\mathbf{Q}$ , such that

$$\langle (E + P) \mathbf{v} \rangle = (\langle E \rangle + \langle P \rangle) \frac{\langle \rho \mathbf{v} \rangle}{\langle \rho \rangle} + \mathbf{Q}_E \quad (5)$$

Let us define the specific total enthalpy,  $h_t \equiv (E + P)/\rho$ . Then we can expand  $\langle (E + P) \mathbf{v} \rangle = \langle \rho \rangle \widetilde{h_t \mathbf{v}}$  and  $\langle E \rangle + \langle P \rangle = \langle \rho \rangle \widetilde{h_t}$ . With this setup, the energy flux can be written as

$$\mathbf{Q}_i = \langle \rho \rangle \left[ \widetilde{h_t v_i} - \widetilde{h_t} \tilde{v}_i \right] = \langle \rho \rangle \tau(h_t, v_i)$$

$h_t$  can be decomposed further into a thermal and kinetic part.

$$h_t = h + \frac{1}{2} v_j v_j \quad h \equiv \frac{\gamma}{\gamma - 1} \frac{P}{\rho}$$

Since  $\tau$  is linear in its first argument,

$$\mathbf{Q}_i = \langle \rho \rangle \tau(h, v_i) + \frac{1}{2} \langle \rho \rangle \tau(v_j v_j, v_i)$$

The first term can be written as

$$\mathbf{Q}_i^{(\text{th})} = \langle \rho \rangle \tau(h, v_i) = \frac{\gamma}{\gamma - 1} [\langle P v_i \rangle - \langle P \rangle \tilde{v}_i]$$

This gives us a thermal/enthalpy flux.

Let's write  $v_j = \tilde{v}_j + \delta v_j$  to expand the second term.

$$v_j v_j = \tilde{v}_j \tilde{v}_j + 2\tilde{v}_j \delta v_j + \delta v_j \delta v_j$$

$\tau(v_j v_j, v_i)$  becomes  $2\tilde{v}_j \tau(v_j, v_i) + \tau(v_j, v_j, v_i)$ , which can be substituted into the kinetic part to get

$$\mathbf{Q}_i^{(\text{kin})} = \langle \rho \rangle [2\tilde{v}_j \tau(v_j, v_i) + \tau(v_j, v_j, v_i)]$$

The first term can be written in terms of the momentum flux tensor, giving us the total energy flux term as

$$\mathbf{Q}_i = \tilde{v}_j \mathbf{R}_{ij} + \frac{1}{2} \langle \rho \rangle \tau(v_j, v_j, v_i) + \langle \rho \rangle \tau(h, v_i) \quad (6)$$

#### The Energy Flux term is not Galilean Invariant

Consider a Galilean transform to  $\mathbf{v} \rightarrow \mathbf{v}' = \mathbf{v} + \mathbf{V}$ .

Under the boost,  $\delta \mathbf{v}$  and all the scalars remain unchanged. But  $\tilde{v}_j \mathbf{R}_{ij}$  term is not going to be invariant. So,

$$\mathbf{Q}'_i = \mathbf{Q}_i + V_j \mathbf{R}_{ij}$$

Finally, we write the emissivity  $\langle \mathcal{E} \rangle$  with a correction term C.

$$\langle n^2 \Lambda(T) \rangle = \left[ \left( \frac{\langle \rho \rangle}{\mu m_p} \right)^2 \Lambda(T) \right] + \mathbf{C} \quad (7)$$

The first term in Equation 7 is the coarse/resolved cooling, and the second term is just how much it is off by, a cooling correction term. Everything we're dealing with here are scalars, so this will be invariant.

## 4 Final Set of Equations

Let us write  $\mathbf{m} = \rho \mathbf{v}$  as the momentum density.

$$\begin{aligned}
\frac{\partial}{\partial t} \langle \rho \rangle + \nabla \cdot \langle \mathbf{m} \rangle &= 0 \\
\frac{\partial}{\partial t} \langle \mathbf{m} \rangle + \nabla \cdot \left( \frac{\langle \mathbf{m} \rangle \langle \mathbf{m} \rangle}{\langle \rho \rangle} + \mathbf{R}_{\rho vv} \right) + \nabla \langle P \rangle &= 0 \\
\frac{\partial}{\partial t} \langle E \rangle + \nabla \cdot \left[ \frac{(\langle E \rangle + \langle P \rangle) \langle \mathbf{m} \rangle}{\langle \rho \rangle} + \mathbf{Q} \right] &= - \left( \frac{\langle \rho \rangle}{\mu m_p} \right)^2 \Lambda(T) - \mathcal{C}
\end{aligned} \tag{8}$$

$\mathbf{R}_{\rho vv}$  has 9 components, but is a symmetric tensor so only 6 independent quantities ( $xx, yy, zz, xy, xz, yz$ ).  $\mathbf{Q}$  will have 3 components.  $\mathcal{C}$  has one component, totalling to 10 unknowns we have to learn.

## 5 Constraining the Cooling term.

From Equation 7, we have the subgrid cooling term  $\mathcal{C}$ . We can also write the resolved cooling given by

$$\mathcal{C}_{CG} = \left( \frac{\langle \rho \rangle}{\mu m_p} \right)^2 \Lambda(T_{CG})$$

which is just the first term in Equation 7.

Figure 1 shows that  $\mathcal{C}$  is only non-zero in the mixing layer, where a coarse cell contains both hot and cold gas. Pure hot or pure cold cells have  $\mathcal{C} = 0$ , because the fine and coarse temperatures then agree. Within the layer,  $\mathcal{C}$  is negative in the upper, hot-side cells and positive in the lower, cold-side cells, so the sign of the correction depends on where the cell sits in the layer.

Figure 2 shows why. On the hot side of the layer,  $\mathcal{C}_{CG}$  is roughly an order of magnitude larger than the true cooling: averaging hot and cold gas puts  $T_{CG}$  in the range where  $\Lambda$  is large, even though little of the gas in the cell is actually at that temperature. On the cold side,  $T_{CG}$  falls below the lower cooling cutoff, so  $\mathcal{C}_{CG} = 0$  even though the cell contains thin filaments of warm gas that are cooling.

Over  $t = 5\text{--}10$  Myr, 17% of the coarse cells have  $\mathcal{C} \neq 0$ , split almost evenly between the two signs (Figure 3). The two distributions are offset, however: the median  $|\mathcal{C}|$  is about 20 times larger for  $\mathcal{C} < 0$  than for  $\mathcal{C} > 0$ . Integrated over the domain,  $\mathcal{C}_{CG}$  therefore overestimates the total cooling by a factor of about 4.6. At the same time, 35% of the cooling cells have  $\mathcal{C}_{CG} = 0$ , and these carry 17% of the true cooling.

Figure 4 shows that the sign of  $\mathcal{C}$  is set mostly by  $T_{CG}$ :

- For  $10^{4.1} < T_{CG} < 10^{5.2}$  K, essentially every cell overcools, with  $|\mathcal{C}| \sim 10^{-24}$  erg cm<sup>-3</sup> s<sup>-1</sup>. The error is largest just above the lower cutoff.
- For  $T_{CG} > 10^{5.5}$  K, essentially every cell undercools.
- Below the lower cutoff,  $\mathcal{C}_{CG} = 0$  and every cell undercools by its full true cooling.
- Between  $10^{5.2}$  and  $10^{5.5}$  K both signs occur in similar numbers, so  $T_{CG}$  alone does not determine  $\mathcal{C}$  there.

A model for  $\mathcal{C}$  therefore needs more than the coarse temperature. It has to know how the gas in the cell is distributed in temperature, which is what the PDF approach for  $\mathcal{C}$  above is designed to provide.

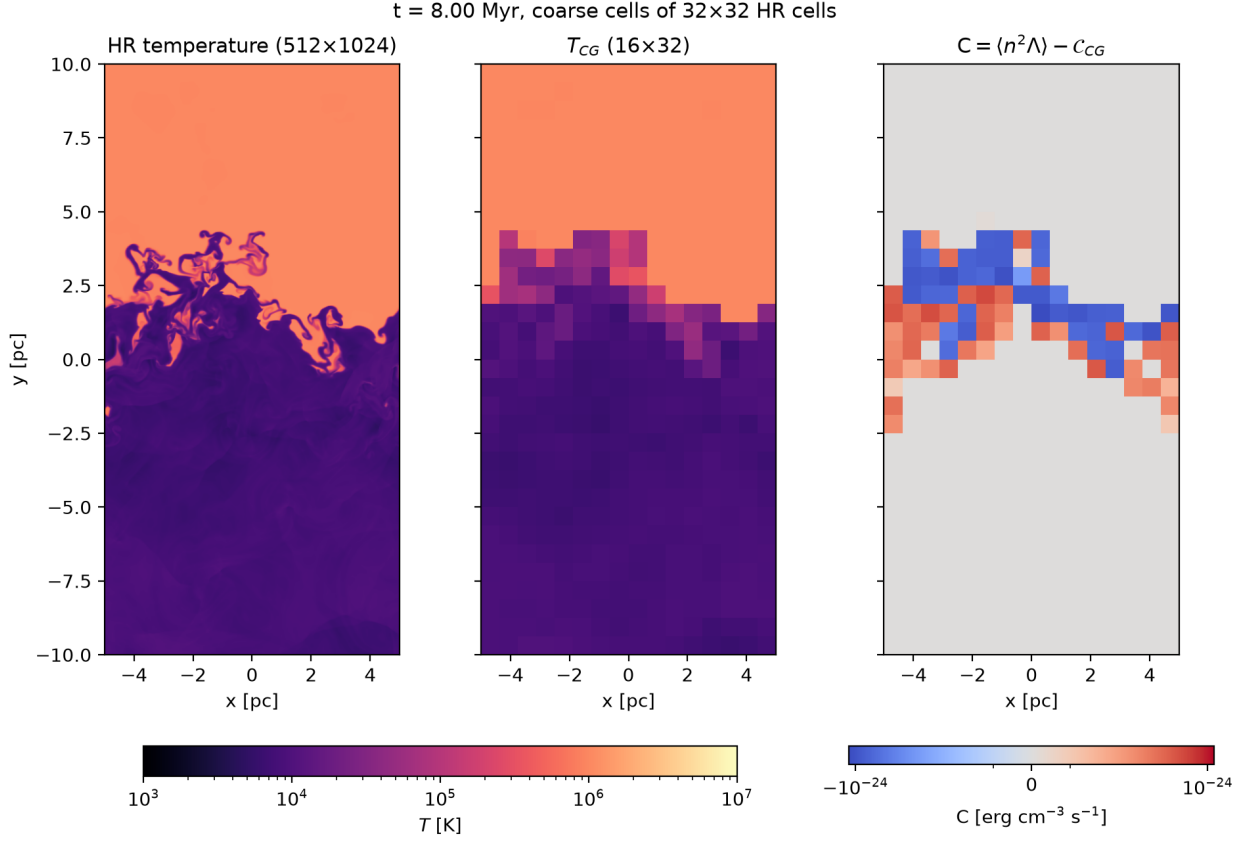


Figure 1: The subgrid cooling term for the 512 × 1024 run at  $t = 8$  Myr, coarse-grained onto blocks of 32 × 32 cells. From left to right: the high-resolution temperature, the coarse temperature  $T_{CG}$ , and the map of  $C = \langle n^2 \Lambda(T) \rangle - C_{CG}$ . Blue cells ( $C < 0$ ) are where  $C_{CG}$  overestimates the cooling; red cells ( $C > 0$ ) are where it underestimates it.

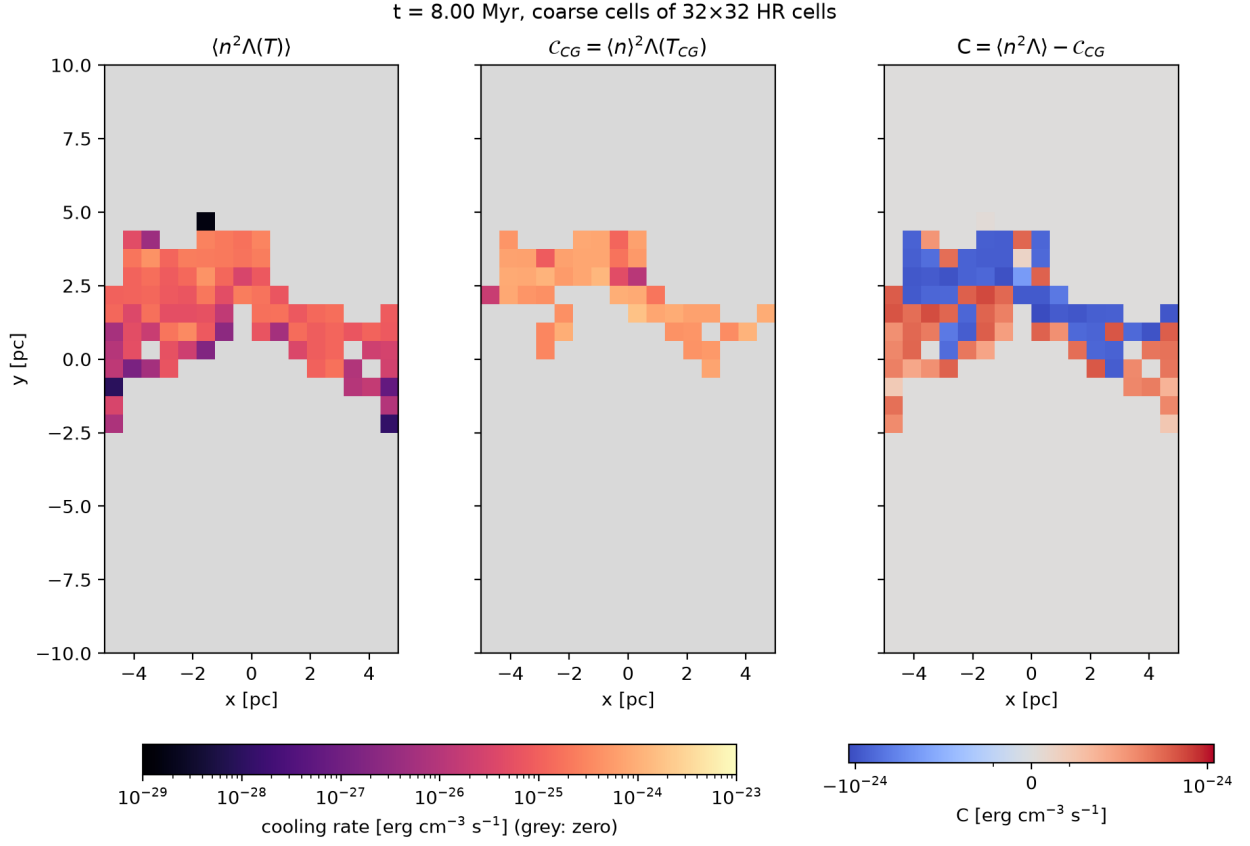


Figure 2: The two cooling estimates and their difference for the same snapshot as Figure 1. From left to right: the true coarse-grained cooling  $\langle n^2 \Lambda(T) \rangle$ , the resolved cooling  $\mathcal{C}_{CG} = \langle n \rangle^2 \Lambda(T_{CG})$  on the same colour scale, and  $C = \langle n^2 \Lambda(T) \rangle - \mathcal{C}_{CG}$ . Grey cells have zero cooling.

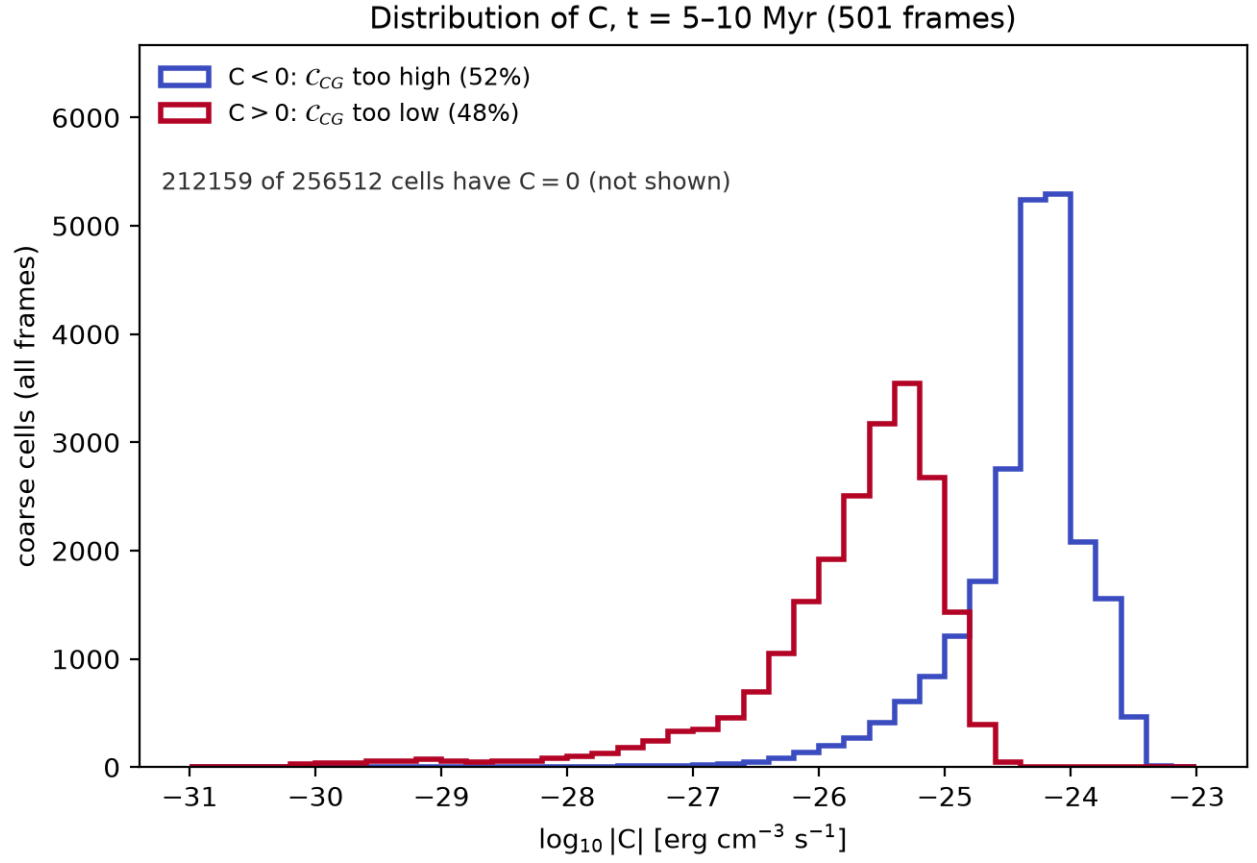


Figure 3: Distribution of  $|C|$  over all coarse cells for  $t = 5\text{--}10$  Myr, split by sign. Cells with  $C = 0$  (pure hot or pure cold gas) are not shown.

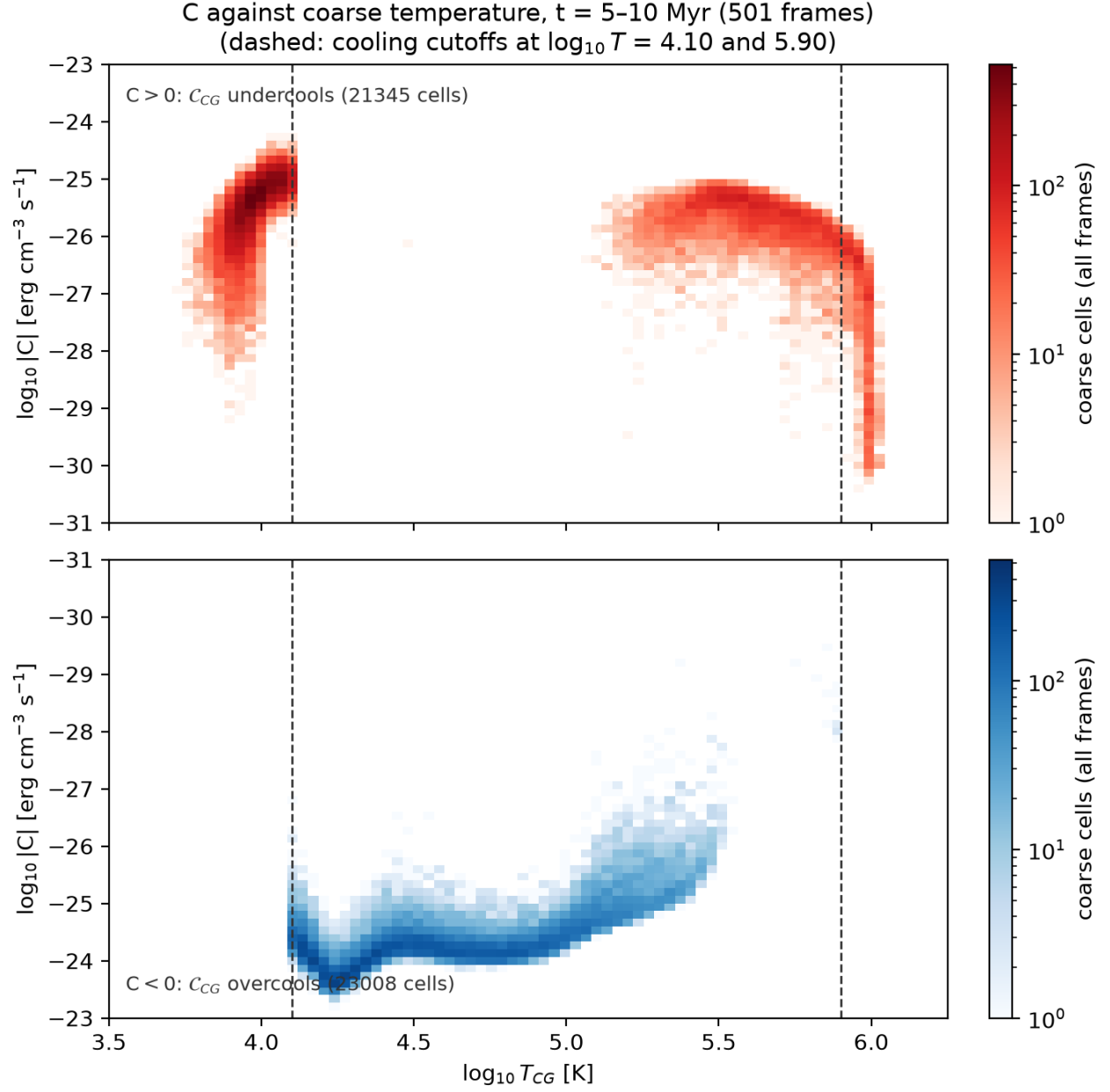


Figure 4: Joint distribution of the coarse temperature  $T_{CG}$  and  $|C|$  for  $t = 5\text{--}10$  Myr. Top: cells where  $C_{CG}$  undercools ( $C > 0$ ). Bottom: cells where it overcools ( $C < 0$ ), with the  $|C|$  axis flipped. Dashed lines mark the cooling cutoffs.

## 5.1 How well does the gate identify cooling cells?

The CNN predicts a gate  $g \in (0, 1)$  for each coarse cell. It switches the predicted temperature PDF between a single sharp peak ( $g \rightarrow 0$ ) and a broad multiphase distribution ( $g \rightarrow 1$ ). The gate is trained against the binary target “the cell’s temperature PDF has mass in the active cooling window”, which is the same as  $\langle n^2 \Lambda(T) \rangle > 0$ . Here we ignore the cooling rate the CNN predicts and only ask how well the gate identifies the cells that should cool. We evaluate the two models used in `run_variable_box_subgrid.sh` (0.625 pc and 1.25 pc cells) on the  $512 \times 1024$  run for  $t = 5\text{--}10$  Myr, coarse-grained to each model’s cell size. Neither model was trained on this run. The figure is produced by `gate_evaluation.py`.

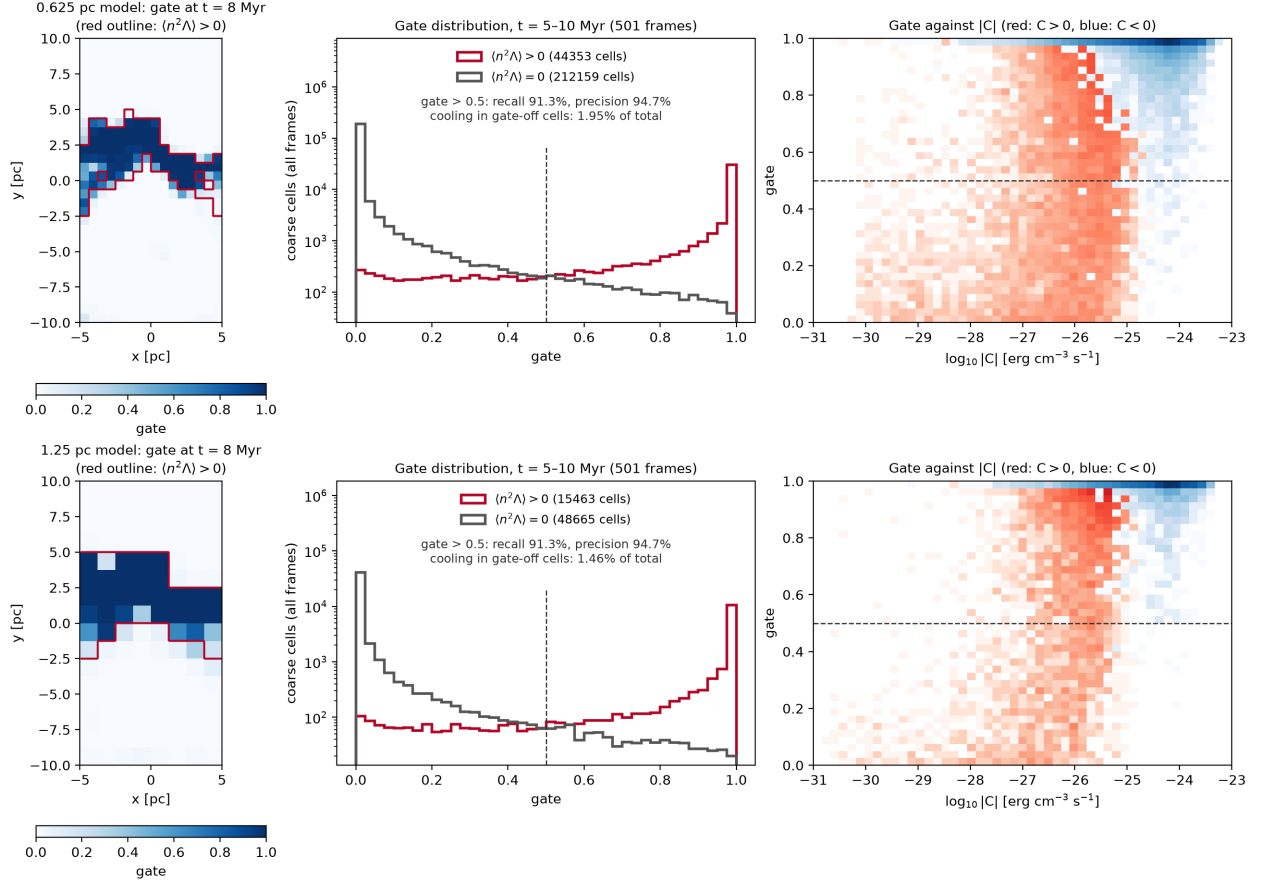


Figure 5: Gate of the 0.625 pc (top) and 1.25 pc (bottom) models. Left: the gate at  $t = 8$  Myr, with the cells where  $\langle n^2 \Lambda \rangle > 0$  outlined in red. Middle: the distribution of the gate for cells that cool and cells that do not, with the threshold  $g = 0.5$  dashed. Right: the gate against  $|C|$  for cells with  $C > 0$  (red) and  $C < 0$  (blue).

Figure 5 shows that the gate works well as a detector of where cooling happens:

- The gate is almost binary. Most cells have  $g < 0.05$  or  $g > 0.95$ , and the two populations separate cleanly at  $g = 0.5$ .
- With  $g > 0.5$  as “on”, both models turn the gate on for 91% of the cooling cells (recall), and 95% of the cells where the gate is on are cooling (precision).
- The cooling cells the gate misses are weak. They carry 2.0% (0.625 pc) and 1.5% (1.25 pc) of the true cooling, and less than 1% of the total  $|C|$ .

- Almost all of the misses (98–99%) are undercooling cells ( $C > 0$ ). About three quarters of them have  $T_{CG}$  below the lower cooling cutoff (median  $T_{CG} \approx 10^{3.95}$  K): cold-dominated cells containing only thin filaments of warm gas. By contrast, the gate is on in more than 99.6% of the overcooling cells ( $C < 0$ ), which carry most of  $|C|$ .

The gate is therefore not the limiting factor for the subgrid cooling. It finds almost all of the cells that matter, and its errors are confined to weakly cooling cells on the cold side of the mixing layer.